

DetectiveNN: Imitating Human Emotional Reasoning with a Recall-Detect-Predict Framework for Emotion Recognition in Conversations

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Abstract

Emotion Recognition in conversations (ERC) involves an internal cognitive process that interprets emotional cues by using a collection of past emotional experiences. However, many existing methods struggle to decipher emotional cues in dialogues since they are insufficient in understanding the rich historical emotional context. In this work, we introduce an innovative Detective Network (DetectiveNN), a novel model that is grounded in the cognitive theory of emotion and utilizes a "recall-detect-predict" framework to imitate human emotional reasoning. This process begins by 'recalling' past interactions of a specific speaker to collect emotional cues. It then 'detects' relevant emotional patterns by interpreting these cues in the context of the ongoing conversation. Finally, it 'predicts' the speaker's current emotional state. Tested on three benchmark datasets, our approach significantly outperforms existing methods. This highlights the advantages of incorporating cognitive factors into deep learning for ERC, enhancing task efficacy and prediction accuracy¹.

1 Introduction

In recent years, recognizing emotions in dialogues has gained increasing attention in the field of natural language processing (NLP). This is driven by its vast potential for application in areas like human-computer interaction and empathetic dialogue systems (Ma et al., 2020; Concannon and Tomalin, 2023).

In the realm of conversational emotion recognition, interpreting emotional cues embedded in conversational context is crucial (Mittal et al., 2020; Gomathy, 2021). Emotional cues in conversations are subtle patterns or indicators that hint at the underlying emotions of a speaker. These cues often act as triggers for the emotions expressed in current utterances (Oberländer et al., 2020; Hu

et al., 2021). ERC seeks to detect and interpret these emotional clues within the flow of conversation, aiming for a nuanced understanding of the emotional context. Traditional ERC approaches typically adopt a 'recall-then-predict' strategy (Mitra et al., 2023), modeling both speaker-level and dialogue-level contexts to predict emotional states in conversations. DialogueGCN (Ghosal et al., 2019) models interactions between speakers using graph networks to capture emotional cues throughout the conversation. DialogXL (Shen et al., 2021) introduces a dialog-aware self-attention mechanism within a transformer structure to capture emotional cues, including intra- and inter-speaker dependencies. C-LSTM (Zhou et al., 2015) leverages an LSTM-based approach to encode the global context, whereas DialogueRNN (Majumder et al., 2019) employs GRUs to track both speaker state and global state for each conversation. COSMIC (Ghosal et al., 2020) leverages external common-sense knowledge to enhance the model's ability to detect rich emotional cues. Additionally, DialogueCRN (Hu et al., 2021) employs a multi-turn reasoning module that extracts and combines emotional clues from the dialogue. However, these methods lack a distinct process for interpreting extracted emotional cues before classifying the emotion.

Emotion recognition can be understood as the process of deciphering emotional cues to comprehend the cognitive context, aligning with the Cognitive Theory of Constructed Emotion (Russell, 2003, 2009; Barrett and Russell, 2014). This theory suggests that emotions are formed from an individual's cognitive context, shaped by their thoughts, memories, and social interactions (Barrett, 2014). Inspired by this theory, we approach ERC tasks as an internal cognitive process that deciphers each participant's emotional cues based on their past emotional experiences in a dialogue. This process involves identifying and organizing emotional cues, synthesizing them into a coherent emotional nar-

¹our code can be found [here](#)

rative, and subsequently examining this narrative throughout the conversational context to validate the cues. We propose a novel Detective Network (DetectiveNN) with a ‘recall-detect-predict’ strategy for enhanced ERC accuracy. The DetectiveNN model features a detection phase that deciphers emotional cues throughout the conversation context, connecting these cues to decode the evolution of a speaker’s emotional responses. This phase reveals patterns in a speaker’s emotional flows, akin to a detective piecing together clues to map an individual’s emotional states.

DetectiveNN begins with a recall phase, where we utilize Gated Recurrent Units (GRUs), a type of sequence-based model, for their efficiency in handling sequential data and their ability to capture long-term dependencies. This is crucial for retaining diverse contextual information from the speakers’ emotional experiences and interactions. This approach is inspired by the pioneering work of Hu (Hu et al., 2021) and Yang (Yang and Shen, 2021), who demonstrated the efficacy of sequence-based models in learning diverse contextual information.

In the detection phase, we employ a transformer-like architecture to iteratively analyze and decode emotional cues drawn from the extensive emotional experiences of a specific speaker. This phase is divided into two key operations: an examination process and a conscious detection process. During the examination phase, we employ transformer encoder blocks with Multi-Head Attention (MHA) layers. MHA layers enable the model to simultaneously focus on various parts of the input sequence by assigning disparate levels of attention or importance. Transformer encoder blocks integrate cues to achieve a deep understanding of the speaker’s emotional context. The conscious detection process employs a cross-attention mechanism, probing the speaker’s constructed emotional narrative and integrating the dynamic interplay between emotional cues and the speaker’s historical interactions. This method uncovers patterns that decode the speaker’s emotional journey, offering insights into the evolution of emotional states over time.

Following the insights gained from the detection phase, an emotion classifier predicts the emotion label of each utterance. By incorporating the ‘recall-detect-predict’ framework, DetectiveNN effectively mirrors the cognitive reasoning process humans use to understand emotional states. We hypothesize that integrating cognitive reasoning into

deep learning models significantly enhances their capability to analyze and interpret emotions in each dialogue segment.

To assess the efficacy of our proposed model, extensive experiments were conducted on three widely accepted benchmark datasets: IEMOCAP, EmoryNLP and Dailydialog. The experimental results demonstrate that our model significantly outperforms existing methods, primarily attributed to the application of a cognitive approach in deciphering emotional cues.

The primary contributions of our research are as follows:

- We introduce an innovative Detective Network (DetectiveNN) designed within a ‘recall-detect-predict’ framework, drawing on principles of cognitive theory of constructed emotion.
- We design a transformer architecture to perform the detection process. This architecture plays a key role in interpreting emotional cues in conversations, enhancing the accuracy and nuances of recognizing different emotions in dialogues.
- We conduct extensive experiments on three benchmark datasets. The results consistently demonstrate the effectiveness and superiority of the proposed model (see Figure 1).

2 Related Work

The ERC field has advanced significantly, emphasizing the extraction and integration of emotional cues from conversations. This progress can be grouped into three major methodologies: Sequence-based models, Pre-trained Language Model-based Models, and Graph-based Models.

Sequence-based models: DialogueRNN (Majumder et al., 2019) uses GRUs to track emotional states by integrating speaker identity, context, and emotions from neighboring utterances. DialogueCRN (Hu et al., 2021) combines cognitive theories of emotion with LSTM networks for iterative extraction and integration of emotional cues. BC-LSTM (Poria et al., 2018) employs bidirectional LSTMs to capture the influence of preceding and following utterances. CMN (Hazarika et al., 2018b) models past utterances using GRUs in a multimodal approach. EmotionIC (Yingjian et al., 2023) integrates attention and recurrence with IMMHA, Di-aGRU, and SkipCRF for comprehensive emotion

detection. COSMIC (Ghosal et al., 2020) incorporates commonsense knowledge with GRUs to model various conversational states.

Pre-trained Language Model-based Models: DialogXL (Shen et al., 2021) uses XLNet (Yang et al., 2019) and dialogue-level self-attention to handle multi-party conversation dynamics. Emoberta (Kim and Vossen, 2021) employs RoBERTa (Liu et al., 2019) to predict speaker emotions by learning both speaker-level and dialogue-level context.

Graph-based Models: DialogueGCN (Ghosal et al., 2019) uses a graph convolutional neural network to model conversational context by representing utterances as nodes and their dependencies as edges. Zhang et al. (2019) and Lian et al. (2020) employ graph convolutional networks with attention mechanisms to capture context-sensitive and speaker-sensitive dependencies.

3 Methodology

3.1 Problem Definition

We define a conversation consisting of a total number of N utterances. Each utterance in the conversation is associated with a specific speaker. There are S distinct speakers in the conversation. For each speaker, we have a subset of utterances corresponding to this speaker.

The objective of the ERC task is to predict the emotion label for each utterance from the set of emotional labels $\{y_1, y_2, \dots, y_P\}$ where P is the number of emotional labels.

3.2 Recall Phase

In the realm of ERC, the intra-context is crucial for understanding the emotional journey and thematic progressions of each speaker within their dialogue contributions.

We first utilize a bi-directional GRU network to gather emotional cues and information from utterances generated by speaker s . Each utterance is represented by a feature embedding $x_i \in \mathbb{R}^{du}$, where du is the embedding dimension of each utterance. The sequence of these embeddings is processed by the GRU, with $i = \Phi(k, s)$ mapping the k -th step in the GRU to the corresponding utterance index for the speaker s .

$$c_i^{\text{intra}}, h_{s,k}^{\text{intra}} = GRU^{\text{intra}}(x_i, h_{s,k-1}^{\text{intra}}) \quad (1)$$

where $c_i^{\text{intra}} \in \mathbb{R}^{2du}$ represents an intra-context embedding, and $h_{s,k}^{\text{intra}}$ is the hidden state of the GRU after processing the k -th step for the speaker s .

We sequentially process each c_i^{intra} and compile them into a matrix $C_s^{\text{intra}} \in \mathbb{R}^{N_s \times 2du}$. N_s is the total number of utterances spoken by the speaker s . This matrix builds up as we go through the steps, eventually leading to the final state.

To obtain the global context embedding c_j^{global} representing all interactions between interlocutors, we employ another bi-directional GRU model to capture sequential dependencies between adjacent utterances of interlocutors. The context representation can be computed as:

$$c_j^{\text{global}}, h_j^{\text{global}} = GRU^{\text{global}}(x_j, h_{j-1}^{\text{global}}) \quad (2)$$

where j is an utterance index from the conversation. Similarly we concatenate c_j^{global} to form the matrix $C^{\text{global}} \in \mathbb{R}^{L \times 2du}$. h_j^{global} is the j -th global hidden state of the GRU.

3.3 Detection Phase

The detection phase offers a systematic method for analyzing the underlying emotional dynamics of the speaker s . Initially, it identifies and organizes emotional cues in a logical order. It then synthesizes those cues to form a coherent emotional narrative. Subsequently, the detection phase examines the emotional narrative against the context of the entire conversation, aiming to validate those initial emotional cues. Throughout this analysis, it uncovers patterns in the emotional flows of the speaker s , akin to a detective connecting dots to reveal a broad map of an individual's emotional states.

Positional Encoding: We first apply positional encoding, denoted as PE, to inject ordering information to the intra-context matrix C_s^{intra} . This ensures that the DetectiveNN not only processes the inherent emotional cues at each step but also understands its sequential context within the entire process.

We adopt transformer encoder blocks with each block consisting of a Multi-Head Attention layer and a Feed-Forward Network layer to identify and integrate emotional cues from the intra-context.

Multi-Head Self-Attention (MHA) Layer: Our architecture includes an MHA layer with four heads to process the intra-context embedding C_s^{intra} . This layer functions as a detective examining the context of speaker s with each head focusing on different aspects of the emotional content in the speaker's utterances. MHA ensures a thorough, multi-faceted analysis by capturing emotional cues from the intra-context.

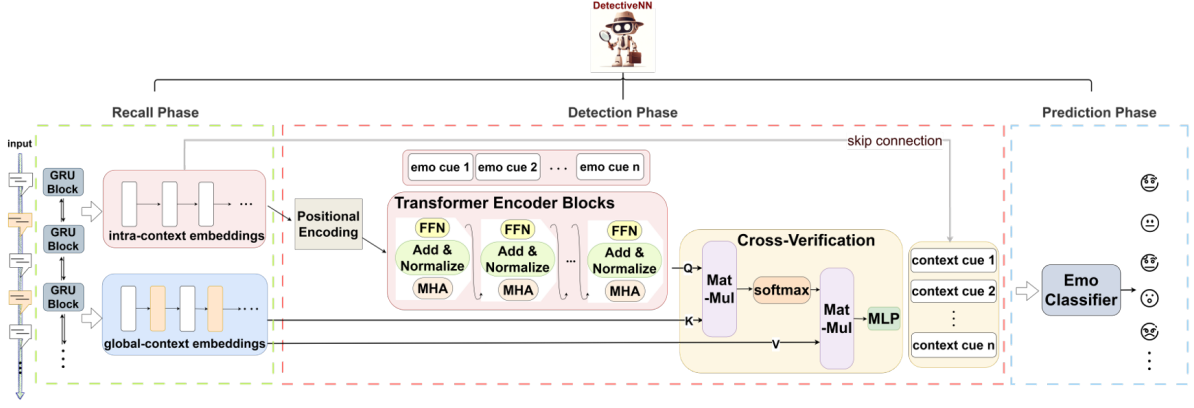


Figure 1: The architecture of the proposed model DetectiveNN

Feed-Forward Network (FFN) Layer: Building on the raw emotional cues identified by the MHA layer, the FFN analyzes how those cues interact and connect. Similar to a detective piecing together different clues in a story, the FFN layer builds a comprehensive emotional narrative of speaker s .

Therefore we obtain $\tilde{C}_s^{\text{intra}}$ as the representation of the emotional narrative. It can be expressed as follows:

$$\tilde{C}_s^{\text{intra}} = \text{FFN}\left(\text{MHA}(C_s^{\text{intra}} + \text{PE}(C_s^{\text{intra}}))\right) \quad (3)$$

where $\tilde{C}_s^{\text{intra}} \in \mathbb{R}^{N_s \times dc}$. dc is the embedding dimension of the emotional narrative.

Cross-Verification Layer: The DetectiveNN then connects emotional cues from intra-context by examining the derived emotional narrative against a broad conversational context. Through careful evaluation, the model identifies patterns in the emotional flows of speaker s . We employ a cross-attention mechanism to mirror this progress. The emotional narrative $\tilde{C}_s^{\text{intra}}$ is treated as a query Q to retrieve additional contextual information from past interactions between the speakers. We set the global context matrix C^{global} as both Key K and Value V .

$$\hat{C}_s^{\text{intra}} = \text{Softmax}\left(\frac{\tilde{C}_s^{\text{intra}} C^{\text{global}T}}{\sqrt{dc}}\right) C^{\text{global}} \quad (4)$$

where $\hat{C}_s^{\text{intra}} \in \mathbb{R}^{N_s \times dc}$ represents emotional patterns captured through cross verification.

3.4 Emotion Prediction

After retrieving and reasoning emotional clues, the detective is to piece together the puzzle in a way to assess the current emotional state of speaker s .

The emotion classification process constitutes the final stage of our model, where we integrate insights derived from the detection phase with a Multi-Layer Perceptron (MLP) layer to predict the emotional state of the targeted utterance.

We employ a skip connection to concatenate original intra-context embedding $c_{i,s}^{\text{intra}}$ with the output of the cross-verification layer $\hat{c}_{i,s}^{\text{intra}}$ along the feature dimension axis. The concatenated feature vector $\mathbb{F}_{i,s}$ represents the updated embedding of the i -th utterance from speaker s :

$$\mathbb{F}_{i,s} = \text{Concat}(\hat{c}_{i,s}^{\text{intra}}, c_{i,s}^{\text{intra}}) \quad (5)$$

Next, $\mathbb{F}_{i,s}$ is fed into the MLP for further processing. The MLP transforms $\mathbb{F}_{i,s}$ into a high-level representation $h_{i,s}$ for making a final prediction:

$$h_{i,s} = \text{MLP}(\mathbb{F}_{i,s}) \quad (6)$$

In the final step, we employ the softmax function to the output of the MLP layer $h_{i,s}$ to obtain a probability distribution over the possible emotional states. The predicted emotional state $\hat{y}_{i,s}$ for the targeted utterance is thus given by:

$$\hat{y}_{i,s} = \text{Softmax}(h_{i,s}) \quad (7)$$

4 Experiments and Results

4.1 Datasets

DetectiveNN was tested on three benchmark datasets for recognizing emotions in conversations: IEMOCAP (Busso et al., 2008), EmoryNLP (Zahiri and Choi, 2018), and DailyDialog (Li et al., 2017). IEMOCAP and DailyDialog focus on two-party dialogues, while EmoryNLP includes multi-party conversations. We report results for all three datasets, with details in Table 1.

Dataset	# Dialogues			# Utterances			Avg. Length	# Classes
	train	val	test	train	val	test		
IEMOCAP	108	12	31	5,810	—	1,623	47	6
DailyDialog	11,118	1,000	1,000	87,832	7,912	7,863	72	7
EmoryNLP	659	89	79	7,551	954	984	10	7

Table 1: Table 1: The statistics of three datasets.

IEMOCAP (Busso et al., 2008): IEMOCAP consists of two-person conversations among ten speakers, with training data from the first eight speakers. Each video captures a dyadic dialogue, divided into utterances annotated with six emotions: happiness, sadness, neutrality, anger, excitement, and frustration.

EmoryNLP (Zahiri and Choi, 2018): EmoryNLP utilizes content from the TV series "Friends," this dataset includes utterances classified into seven emotions: neutral, joyful, peaceful, powerful, scared, mad, and sad. Sentiments are labeled as positive, negative, or neutral.

DailyDialog (Li et al., 2017): DailyDialog covers various everyday topics, mirroring natural human conversation. Each utterance is annotated with emotional categories and dialogue acts, including seven emotions: angry, disgusted, fearful, joyful, neutral, sad, and surprised.

Our research primarily investigates the emotional categorization and text aspects of these datasets. We align our study with COSMIC’s (Ghosal et al., 2020) train/validation/test splits for consistency.

4.2 Baselines

We compare our model, DetectiveNN, with several models introduced in the related work section, including DialogueRNN, DialogueGCN, DialogueCRN, BC-LSTM, CMN, EmotionIC, COSMIC, and DialogXL. Additionally, we also evaluate DetectiveNN against two other models: EmoCaps and CNN.

EmoCaps (Li et al., 2022): EmoCaps utilizes a transformer-based architecture to extract emotional trends across various modalities. It leverages a bi-directional LSTM for contextual analysis, integrating both past and future conversational context to classify emotions.

CNN (Kim, 2014): CNN is a convolutional neural network designed to be trained on utterances that are context-independent.

Table 2, Table 3, and Table 4 present the perfor-

mance evaluation of DetectiveNN on the test data. In training the model on the IEMOCAP dataset, we integrate textual, visual and audio features to create multimodal fused embeddings. All three modality feature embeddings are obtained from Li et al. (2022). For training the model on the EmoryNLP and DailyDialog datasets, we utilized RoBERTa to extract contextual features. RoBERTa embeddings are taken from Ghosal et al. (2020).

4.3 Evaluation Metrics

Consistent with prior studies by Hazarika et al. (2018a), Ghosal et al. (2020), and Jiao et al. (2020), we select the accuracy score (Acc.) as our primary metric for evaluating overall performance on the IEMOCAP, EmoryNLP, and DailyDialog datasets. Additionally, to provide a comprehensive assessment of our model’s capability across both majority and minority classes, we report both the Weighted-average F1 score (Weighted-F1) and the Macro-averaged F1 score (Macro-F1) for IEMOCAP and EmoryNLP datasets. We report both the micro-average F1 score (Micro-F1) and the Macro-averaged F1 score (Macro-F1) for the DailyDialog dataset. These metrics offer a more nuanced view of the model’s effectiveness in handling different class distributions.

4.4 Implementation Details

In our experimental setup, the validation set is utilized for hyperparameter optimization. The architecture varies between datasets: a single-layer bidirectional GRU is applied to IEMOCAP, EmoryNLP and Dailydialog datasets.

In the subsequent detection phase, we employed two transformer encoder blocks because the EmoryNLP dataset is characterized by a limited number of turns and brief conversations. This configuration facilitates a more prolonged learning process, allowing the model to effectively detect nuanced emotional cues within short conversational contexts. Our experiments also demonstrate that incorporating additional encoder blocks en-

ables the model to identify a broader range of features. For the IEMOCAP and Dailydialog datasets, which are characterized by longer turns and more extended conversations, we implemented a single transformer encoder block.

The batch size is uniformly maintained at 30 for all experiments. For each dataset, the learning rate and dropout are set specifically as follows: 10^{-3} and 0.5 for IEMOCAP, 10^{-4} and 0.2 for EmoryNLP, and 10^{-4} and 0.5 for Dailydialog. L2 weight decay is set to 2×10^{-3} for all experiments. The loss objective for all experiments is cross-entropy loss. We trained the DetectiveNN for a maximum of 80 epochs using the Adam optimizer (Kingma and Ba, 2014) and stopped training if the validation loss did not decrease for 10 consecutive epochs. For benchmarking against existing models like CNN, BC-LSTM, DialogueGCN, DialogueRNN, and DialogueCRN, we replicate their setups using the publicly available code provided by Kim (2014), Poria et al. (2018), Majumder et al. (2019), Ghosal et al. (2019), and Hu et al. (2021), ensuring consistency in the experimental environment.

Methods	Acc.	Weighted-F1	Macro-F1
CNN †	53.16	52.13	47.28
BC-LSTM †	55.86	55.24	53.19
CMN*	56.56	56.13	54.30
COSMIC*	—	65.28	—
DialogXL*	—	65.94	—
DialogueRNN†	63.50	63.18	62.99
DialogueGCN†	62.42	62.11	61.17
DialogueCRN†	70.65	70.35	70.01
Emoberta*	—	68.57	—
EmoCaps*	—	71.77	—
DetectiveNN	76.15	76.01	76.40

Table 2: Experimental results on the IEMOCAP dataset. Annotated with an * indicates results sourced from the model’s paper, and a (†) denotes results from reproductions conducted by the authors.

Methods	Acc.	Micro-F1	Macro-F1
CNN†	65.35	57.21	50.13
BC-LSTM†	64.19	53.19	48.94
EmotionIC*	—	60.13	54.19
COSMIC*	—	58.48	51.05
DialogXL*	—	54.93	—
DialogueRNN†	63.03	61.50	57.66
DialogueGCN†	71.56	62.20	60.43
DialogueCRN†	73.15	64.10	53.18
DetectiveNN	75.55	70.20	57.38

Table 3: Experimental results on the Dailydialog dataset. Annotated with an * indicates results sourced from the model’s paper, and a (†) denotes results from reproductions conducted by the authors.

Methods	Acc.	Weighted-F1	Macro-F1
CNN†	34.21	30.19	28.59
BC-LSTM†	38.17	34.27	29.87
SACL-LSTM*	—	39.65	—
COSMIC*	—	38.11	—
DialogXL*	—	34.73	—
DialogueGCN†	37.75	34.98	31.30
DialogueCRN†	40.65	37.59	32.31
DialogueRNN†	41.04	35.76	31.22
EmotionIC*	—	40.25	—
DetectiveNN	42.68	40.78	33.65

Table 4: Experimental results on the EmoryNLP dataset. Annotated with an * indicates results sourced from the model’s paper, and a (†) denotes results from reproductions conducted by the authors.

4.5 Main Results

Table 2, Table 3, and Table 4 illustrate the results of comparing our DetectiveNN model with other models and backbones from different perspectives. Based on this, we make the following observations:

(1) Our method achieves significant improvements over the SOTA baseline models on all benchmarks. Specifically, we outperform EmoCaps, DialogueCRN, and EmotionIC by 4.24%, 6.10%, and 0.53% on IEMOCAP, Dailydialog and EmoryNLP respectively.

(2) Previous research has highlighted the complexity involved in emotion modeling in the EmoryNLP dataset, challenges stemming from the diversity of speakers and limited conversational exchanges (Ghosal et al., 2019; Li et al., 2020). DetectiveNN, in contrast, shows notable performance enhancements on the IEMOCAP and DailyDialog datasets. This advancement is attributed to longer and more in-depth conversational exchanges and richer utterance content in these datasets. These aspects allow for a more comprehensive understanding of the global context and emotional cues, thus enhancing the accuracy of DetectiveNN.

4.6 Ablation Study

The DetectiveNN model is based on a recall-detect-predict framework. To assess the impact of the recall and detection phases on performance, ablation experiments were conducted on the IEMOCAP and EmoryNLP datasets. Removing either phase successively led to a significant performance drop, demonstrating their importance. Detailed results are in Table 5.

Recall Phase Analysis: The recall phase gathers relevant global context from dialogues. Excluding this phase reduced the model’s effectiveness

Context	Cognition		IEMOCAP			EmoryNLP		
Intra-Contextual Dependency	Recall Phase	Detection Phase	Acc.	W-F1	M-F1	Acc.	W-F1	M-F1
✓	✓	✓	76.15	76.01	76.40	42.68	40.78	33.65
✓	✓	×	51.60	50.38	50.62	38.21	36.03	29.15
✓	×	✓	41.46	38.60	36.77	39.11	37.39	31.15
✓	×	×	70.40	70.68	70.98	40.55	38.35	30.85
×	✓	✓	57.74	57.13	57.80	37.60	37.10	30.45
×	✓	×	50.26	50.14	50.30	38.92	37.00	30.07

Table 5: Experimental results of ablation studies on IEMOCAP and EmoryNLP datasets.

on both datasets, highlighting its crucial role in forming a contextual base for reasoning.

Detection Phase Analysis: The detection phase analyzes emotional cues from the recall phase. The absence of this phase resulted in a marked decrease in performance across both datasets, as highlighted in our results. This decline shows the critical role of the detection phase in decoding emotional cues in a conversational context. Furthermore, our findings, as detailed in the final row of Table 5, reveal that eliminating both the recall and detection phases also results in a significant drop in performance. This marked decline highlights the interdependent and synergistic nature of these two phases, underlining their combined importance in augmenting the reasoning capability of the DetectiveNN model.

Impact of Intra-Contextual Dependency: Our study further explored the significance of intra-contextual dependency, essential for understanding how a speaker’s emotional state is shaped by their unique conversational context. Excluding this dependency-tracking component from DetectiveNN resulted in a great decline in performance across both datasets. This outcome highlights the imperative for DetectiveNN to effectively monitor each speaker’s emotional journey, allowing the model to accurately identify and interpret personal emotional cues.

4.7 Comparative Case Study

We also conducted a comparative case study to evaluate our method against the DialogueCRN and DialogueRNN models. Table 6 shows a conversation sampled from the IEMOCAP dataset. DialogueRNN failed to capture the complex emotional context in utterances, such as mislabeling "I want you to pretend like he’s coming back!" as "sad," which misses underlying frustration and possibly anger. Similarly, it incorrectly predicted "excited"

for "But, Kate....," showing a lack of understanding of emotional nuances and situational context. DialogueCRN also demonstrated limitations, such as misinterpreting conflict when it incorrectly labeled the utterance "Laugh at me, but what happens the night that she goes to sleep in his bed, and his memorial breaks in pieces?" as "sad."

In contrast, our model, DetectiveNN, employs a recall-detect-predict framework that demonstrates more accurate emotion recognition in dialogues. In this case, the predicted labels from DetectiveNN alternate between “frustrated” and “angry,” while the predictions of DialogueCRN and DialogueRNN include more varied and less stable labels. The results suggest that DetectiveNN takes better advantage of historical information, meaning that it respects emotional inertia.

5 Visualization

We present heatmap visualizations to interpret the model’s emotion predictions. We randomly selected a conversation sample from the IEMOCAP test dataset for visualization. The first heatmap (see Figure 2) shows features from the transformer encoder’s final layer during the detection phase. The second heatmap (see Figure 3) displays the output of the MLP’s first layer. In both heatmaps, the x-axis of the heatmap represents the dimensions of the feature vector, with each value derived from the mean feature values for each emotion class. The y-axis represents different emotion classes.

The first heatmap reveals distinct activation patterns across the 200 feature dimensions, varying significantly between emotion classes (e.g., angry, sad). These patterns may capture emotional cues such as certain lexical features associated with predicting ‘sad’ or ‘angry’ classes. However, the model struggles to differentiate closely related negative emotions, like ‘angry’ and ‘frustrated.’ This

Ground-truth label	Detective NN	Dialogue CRN	Dialogue RNN	A Case Study
frustrated	frustrated	frustrated	neutral	Person B: Look. It's a nice day. Why are we arguing?
frustrated	frustrated	angry	angry	Person A: Nobody in her—this house dares shake her faith. Strangers might, but not his father, and not his brother.
frustrated	frustrated	neutral	frustrated	Person B: What do you want me to do? What do you want—
angry	angry	frustrated	sad	Person A: I want you to pretend like he's coming back!
frustrated	frustrated	neutral	excited	Person B: But, Kate...
angry	angry	excited	angry	Person A: Because if he's not coming back, then I'll kill myself.
frustrated	frustrated	frustrated	excited	Person B: Hah.....
angry	angry	sad	angry	Person A: Laugh at me, but what happens the night that she goes to sleep in his bed, and his memorial breaks in pieces?

Table 6: A comparative case study.

may be due to the insufficiency of emotional cues alone for interpreting these granular differences. The second heatmap (see Figure 3) shows more intense activations for the ‘frustrated’ class than the ‘angry’ class. These patterns suggest that the model captures and integrates contextual information to decode the complex interplay between emotions, specific events, and the unfolding narrative within a sentence.

6 Conclusions

In this paper, we introduce DetectiveNN, a novel framework for Emotion Recognition in Conversation. This framework utilizes an innovative recall-detect-predict structure to interpret emotions in conversations. Initially, DetectiveNN identifies key emotional cues within the dialogue. Subsequently, it conducts a thorough analysis of these cues to accurately predict the emotional state.

Rigorously evaluated across three benchmark datasets, DetectiveNN has demonstrated its superiority over existing models, revealing the profound impact of integrating cognitive reasoning into deep learning architectures. This cognitive factor plays an important role not only in enhancing the model’s efficiency and accuracy in prediction but also in advancing ERC methodologies.

7 Limitations

DetectiveNN improves emotion prediction accuracy for long-term dialogue turns but struggles with short-term turns due to its reliance on extended interaction context. This context dependence limits its ability to detect emotional cues in brief exchanges. Additionally, lacking information on speakers’ personality traits hinders DetectiveNN’s performance in capturing complex emotional dynamics, such as sarcasm and humor, which

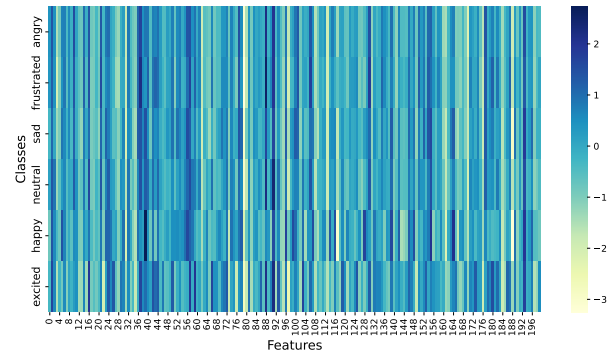


Figure 2: Output of the last layer of the transformer encoder block.

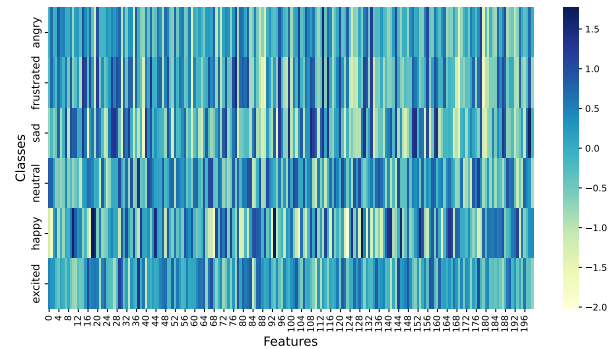


Figure 3: Output of the first layer of MLP layer.

are prevalent in datasets like EmoryNLP from the "Friends" TV series. Integrating personality trait knowledge is essential for accurately predicting nuanced emotions in conversations.

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